

AI-Powered Cybersecurity Prompt Library

A professional collection of practical AI prompts for cybersecurity students, SOC analysts, ethical hackers, blue teamers, and IT learners.

Best Practices for Using AI in Cybersecurity

- Sanitize sensitive data before sharing logs or incidents
- Remove passwords, API keys, and confidential identifiers
- Validate AI outputs manually
- Use AI for explanations, summaries, detection logic, and workflow acceleration
- Never rely entirely on AI for final incident attribution or remediation decisions

Limitations of AI in Cybersecurity

- AI can hallucinate indicators or misclassify benign activity
- AI may oversimplify attack chains
- AI-generated code may contain vulnerabilities
- Human validation is always required

1. Vulnerability Explanation Prompts

SQL Injection Prompt:
Act as a cybersecurity instructor.
Explain the following SQL Injection vulnerability in beginner-friendly terms.

[INSERT DETAILS]

Include:

- Attack flow
- Risk explanation
- Mitigation guidance
- Detection opportunities
- Real-world examples

XSS Prompt:
Explain Stored, Reflected, and DOM-based XSS.
Include browser impact, mitigation, CSP guidance, and secure coding practices.

CSRF Prompt:
Explain how CSRF works, why authenticated users are targeted, and how CSRF tokens help.

SSRF Prompt:
Explain SSRF risks including cloud metadata exposure and internal network targeting.

Broken Authentication Prompt:
Explain authentication weaknesses, account takeover risks, and MFA recommendations.

Privilege Escalation Prompt:
Explain vertical vs horizontal escalation and detection opportunities.

RCE Prompt:
Explain remote code execution risks and ransomware connections.

Directory Traversal Prompt:
Explain path traversal attacks and secure file handling practices.

2. SOC Analyst & Log Analysis Prompts

Act as a SOC analyst.
Analyze the following logs and identify:

- Suspicious activity
- Indicators of compromise
- MITRE ATT&CK techniques
- Severity level
- Recommended next steps

Logs:
[INSERT LOGS]

Failed Login Analysis:
Detect brute force attempts, password spraying, and suspicious IP activity.

PowerShell Detection:
Identify encoded commands, persistence mechanisms, and obfuscation attempts.

Firewall Log Analysis:
Detect scanning behavior, suspicious outbound traffic, and C2 indicators.

3. Incident Response Prompts

Ransomware Response:
Create an incident response checklist for a ransomware infection affecting Windows endpoints.

Include:

- Triage
- Containment
- Evidence preservation
- Recovery workflow
- Communication guidance

Phishing Investigation:
Analyze phishing emails and extract suspicious domains and credential theft indicators.

Insider Threat Investigation:
Create a workflow for investigating suspicious employee behavior and access abuse.

4. Threat Hunting Prompts

Analyze endpoint activity for:

- Persistence mechanisms
- Registry abuse

- Scheduled tasks
- Service creation

Lateral Movement Detection:

Identify pass-the-hash behavior, RDP abuse, SMB anomalies, and credential reuse.

Network Anomaly Detection:

Identify beaconing, DNS tunneling, and unusual outbound connections.

5. OSINT & Reconnaissance Prompts

Analyze DNS records and identify:

- Exposed services
- Misconfigurations
- Attack surface risks

WHOIS Investigation:

Identify suspicious domain registration patterns and phishing indicators.

Attack Surface Mapping:

Analyze public-facing assets for exposure and risky configurations.

6. Malware Analysis Prompts

Analyze suspicious scripts and explain:

- What the script does
- Persistence mechanisms
- Network activity
- IOC extraction
- Detection opportunities

Obfuscated Code Prompt:

Explain suspicious functionality and malware techniques.

Sandbox Analysis:

Interpret sandbox results including registry changes and process injection.

7. Cybersecurity Study Assistant Prompts

Generate:

- Flashcards
- Practice quizzes
- Exam prep questions
- Simplified technical explanations
- Progressive CTF hints

Example:

Generate 20 beginner-to-intermediate cybersecurity quiz questions covering networking, Linux, and web security

8. Report Writing & Documentation Prompts

Generate:

- Vulnerability reports
- Executive summaries
- SOC incident reports
- Technical documentation

Include:

- Severity ratings
- Remediation guidance
- Business impact
- Executive-friendly summaries

9. Automation & Scripting Prompts

Generate beginner-friendly scripts for:

- Log parsing
- IOC extraction
- Port scanning
- Authentication monitoring

Requirements:

- Add comments
- Include error handling
- Explain the code
- Use clean coding practices

10. Blue Team Defensive AI Workflows

Alert Prioritization:

Prioritize alerts based on severity, likelihood, and business impact.

Detection Engineering:

Create Sigma logic and SIEM queries for suspicious behavior.

Threat Intelligence Summarization:

Extract TTPs, IOCs, and defensive recommendations from threat reports.

Security Awareness Content:

Generate user-friendly security awareness material.

Reusable Prompt Engineering Structure

Act as a [ROLE].

Analyze the following [DATA TYPE]:

[INSERT DATA]

Objectives:

- Identify risks
- Explain findings
- Recommend defenses
- Assess severity

Output:

1. Summary
2. Key Findings
3. Risk Assessment

4. Recommendations
5. Detection Opportunities

Verification Reminder

Always validate AI-generated analysis manually.

Recommended validation methods:

- SIEM correlation
- Sandbox testing
- Packet analysis
- Threat intelligence verification
- Manual review
- EDR/XDR validation

AI accelerates workflows but does not replace cybersecurity judgment.